

connections:

ESP8266 WiFi Module

Connected to Arduino via Software Serial

RX Pin: Connected to Arduino digital pin 11

TX Pin: Connected to Arduino digital pin 10

EN:3.3v supply (not explicitly shown in code but required)

3.3V: Needs 3.3V supply (not explicitly shown in code but required)

GND: Connected to Arduino GND

Ultrasonic Sensor (HC-SR04)

Trigger Pin: Connected to Arduino digital pin 2

Echo Pin: Connected to Arduino digital pin 3

VCC: Connected to Arduino 5V (not explicitly shown but required)

GND: Connected to Arduino GND (not explicitly shown but required)

LCD Display (16x2)

Using standard 16-pin connection mode with the following pins:

RS (Register Select): Connected to Arduino analog pin A0

EN (Enable): Connected to Arduino analog pin A1

D4 (Data pin 4): Connected to Arduino analog pin A2

D5 (Data pin 5): Connected to Arduino analog pin A3

D6 (Data pin 6): Connected to Arduino analog pin A4

D7 (Data pin 7): Connected to Arduino analog pin A5

VSS: Connected to Arduino GND (not explicitly shown but required)

VDD: Connected to Arduino 5V (not explicitly shown but required)

V0: Connected to a potentiometer for contrast control (not explicitly shown but required)

RW: Typically connected to GND for write-only mode (not explicitly shown)

A: Connected to Arduino 5V via a resistor (not explicitly shown but required)

K: Connected to Arduino GND (not explicitly shown but required)

Servo Motor

Yellow/Orange(Signal): Connected to Arduino digital pin 9

Red(VCC): Connected to external power supply or Arduino 5V (not explicitly shown but required)

Black/Brown(GND): Connected to Arduino GND (not explicitly shown but required)

LEDs

Red LED: +ve leg Connected to Arduino digital pin 4 , -ve leg to GND

Green LED: +ve leg Connected to Arduino digital pin 5 , -ve leg to GND

Buzzer

Positive leg: Connected to Arduino digital pin 7

Negative leg: Connected to Arduino GND (not explicitly shown but required)

don't forget to connect potentiometer to v5 and GND